CONTINUOUS REINFORCEMENT LEARNING

Abstract.

1. Introduction

2. Continuous approach of the problem

We consider an optimal control problem framed within a continuous state space and over continuous time. The framework models a deterministic system subject to discounted future rewards.

The evolution of the system's state, denoted by $x(t)$ at time t, is driven by a control input $u(t)$. We establish the following topological and algebraic conditions:

(1) $x(t) \in \Omega \subset \mathbb{R}^d$ denotes the state of the system at time t, where Ω is an open subset representing the state space.
(2) $u(t) \in \mathcal{U} \subset \mathbb{R}^m$ denotes the control input at time t, where the action space $\mathcal{U}$ is assumed to be a compact subset to ensure the existence of optimal control policies.

Definition 2.1 (System Dynamics)**.** The time evolution of the system is governed by the following first-order ordinary differential equation:

$$\dot{x}(t) = f(x(t), u(t)), \quad x(0) = x_0$$

where $f : \Omega \times \mathcal{U} \to \mathbb{R}^d$ describes the vector field dictating the state transition. To guarantee the well-posedness of the system, we assume:

(1) f is continuous on $\Omega \times \mathcal{U}$.
(2) f is uniformly Lipschitz continuous with respect to the state variable x, meaning there exists a constant $L > 0$ such that $\|f(x, u) - f(y, u)\| \leq L\|x - y\|$ for all $x, y \in \Omega$ and $u \in \mathcal{U}$.

Consequently, by the Picard-Lindelöf theorem, any choice of an initial state $x_0 \in \Omega$ and an admissible, Lebesgue-measurable control function $u : [0, \infty) \to \mathcal{U}$ determines a unique, absolutely continuous state trajectory $x(t)$ for $t \geq 0$.

Definition 2.2 (Performance Functional)**.** Let $\tau = \inf\{t \geq 0 \mid x(t) \notin \Omega\}$ denote the first exit time of the state trajectory from the open domain Ω. We define the performance functional $J(x; u(\cdot))$ as the *discounted cumulative reinforcement*:

$$J(x; u(\cdot)) = \int_0^\tau \gamma^t r(x(t), u(t))dt + \gamma^\tau R(x(\tau))$$

To guarantee that the optimal control problem is mathematically well-posed, we impose the following fundamental assumptions:

(1) **Discounting:** The discount factor satisfies $\gamma \in (0, 1)$, ensuring the convergence of the functional over long or infinite horizons.
(2) **Boundedness and Continuity:** The system dynamics f, the running reward r, and the terminal reward R are uniformly bounded and uniformly Lipschitz continuous with respect to the state variable x. This bounds the maximum possible reinforcement and strictly controls the error propagation over time.
(3) **Compactness of the Action Space:** The control set $\mathcal{U} \subset \mathbb{R}^m$ is compact, this ensures that the supremum evaluated in the value function is attainable, guaranteeing the existence of an optimal control action.

(4) **Domain Regularity:** The boundary of the state space, $\partial\Omega$, is of class $\mathcal{C}^2$. This geometric smoothness prevents singularities in the exit time τ, thereby ensuring that the value function remains sufficiently differentiable to satisfy the Hamilton-Jacobi-Bellman, Theorem **??**.

Theorem 2.3 (*Dynamic Programming Principle*)**.** *Let $f : \Omega \times U \to \mathbb{R}^d$ define the dynamics of a deterministic control system:*

$$\dot{x}(t) = f(x(t), u(t)), \quad x(0) = x$$

where $U \subset \mathbb{R}^m$ is a compact action space, and $\mathcal{U}$ is the set of all admissible, Lebesgue-measurable control functions $u : [0, \infty) \to U$. Let $r : \Omega \times U \to \mathbb{R}$ be a bounded reward function, and $\gamma \in (0, 1)$ be the discount factor. Define the value function as the supremum over all admissible controls:

$$V(x) = \sup_{u \in \mathcal{U}} \left\{ \int_0^\tau \gamma^t r(x(t), u(t))dt + \gamma^\tau R(x(\tau)) \right\}$$

Then, for any sufficiently small time step $h > 0$ such that $h < \inf_{u \in \mathcal{U}} \tau(x, u)$, the following recursive identity holds:

$$V(x) = \sup_{u \in \mathcal{U}} \left\{ \int_0^h \gamma^t r(x(t), u(t))dt + \gamma^h V(x(h)) \right\}$$

where $x(t)$ is the trajectory of the system under control $u(t)$.

Proof. Let $\mathcal{U}$ denote the set of all admissible control functions. Given any control $u \in \mathcal{U}$, we can strictly decompose it into a concatenation of two independent control functions at time h:

$$u(t) = \begin{cases} u_1(t), & \text{for } t \in [0, h), \\ u_2(t - h), & \text{for } t \geq h \end{cases}$$

where $u_1(t) = u(t)$ for $t \in [0, h)$, and $u_2(s) = u(s+h)$ for $s \geq 0$. Let $x(t)$ denote the state trajectory starting from x under $u(t)$. Because we assume $h < \inf_{u \in \mathcal{U}} \tau(x, u)$, the cost functional naturally decomposes into two intervals without exiting the domain Ω:

$$J(x; u) = \int_0^h \gamma^t r(x(t), u_1(t))dt + \int_h^\tau \gamma^t r(x(t), u_2(t - h))dt + \gamma^\tau R(x(\tau))$$

Applying the change of variables $s = t - h$ to the second integral, the integration bounds shift to $[0, \tau - h]$, yielding:

$$J(x; u) = \int_0^h \gamma^t r(x(t), u_1(t))dt + \gamma^h \int_0^{\tau - h} \gamma^s r(x(h + s), u_2(s))ds + \gamma^\tau R(x(\tau))$$

The last two terms can be factored as follows:

$$J(x; u) = \int_0^h \gamma^t r(x(t), u_1(t))dt + \gamma^h \left(\int_0^{\tau - h} \gamma^s r(x(h + s), u_2(s))ds + \gamma^{\tau - h} R(x(\tau)) \right)$$

Observe the mathematical structure of the term inside the parenthesis:

i. The state $x(h)$ acts as the new initial condition for the remaining trajectory, reached strictly by the application of control u_1 over $[0, h)$. We denote this intermediate state as $x_{u_1}(h)$.
ii. The expression inside the parenthesis is exactly the cost functional evaluated from this new initial state using the shifted control u_2. Therefore, it equals $J(x_{u_1}(h); u_2)$.

Substituting this back, we obtain the decomposed cost:

$$J(x;u) = \int_0^h \gamma^t r(x(t), u_1(t))dt + \gamma^h J(x_{u_1}(h); u_2)$$

The supremum over all admissible controls $u \in \mathcal{U}$ is equivalent to taking the independent suprema over all possible initial segments u_1 and all possible tail segments u_2:

$$V(x) = \sup_{u_1, u_2 \in \mathcal{U}} \left\{ \int_0^h \gamma^t r(x(t), u_1(t))dt + \gamma^h J(x_{u_1}(h); u_2) \right\}$$

Since the choice of u_2 does not affect the integral over $[0, h)$, we can isolate the inner supremum:

$$V(x) = \sup_{u_1 \in \mathcal{U}} \left\{ \int_0^h \gamma^t r(x(t), u_1(t))dt + \gamma^h \sup_{u_2 \in \mathcal{U}} \left\{ J(x_{u_1}(h); u_2) \right\} \right\}$$

By definition, $\sup_{u_2 \in \mathcal{U}}\{J(x_{u_1}(h); u_2)\}$ is exactly the optimal value function evaluated at the intermediate state, $V(x_{u_1}(h))$. Reverting the notation $u_1(t)$ back to the generic control $u(t)$ for the interval $[0, h)$, we arrive at the Bellman Principle of Optimality:

$$V(x) = \sup_{u \in \mathcal{U}} \left\{ \int_0^h \gamma^t r(x(t), u(t))dt + \gamma^h V(x(h)) \right\}$$

□

Lemma 2.4. *Let $r(x, u)$ be a bounded reward function that is Lipschitz continuous in x, uniformly with respect to u. Assume $x(t)$ is a state trajectory with initial condition $x(0) = x$, satisfying $x(t) = x + O(t)$[1] as $t \to 0$.*

Then, for a sufficiently small time step $h > 0$, the integrand can be approximated as $r(x(t), u) = r(x, u) + O(t)$ uniformly for $t \in [0, h]$, and the discounted integral satisfies:

$$\int_0^h \gamma^t r(x(t), u)\, dt = r(x, u) \cdot h + O(h^2)$$

Proof. By the uniform Lipschitz continuity of r with respect to x, there exists a constant $L > 0$ such that for any control u:

$$|r(x(t), u) - r(x, u)| \leq L\|x(t) - x\|$$

Since the trajectory satisfies $\|x(t) - x\| = O(t)$, it follows that:

$$|r(x(t), u) - r(x, u)| = O(t)$$

uniformly for $t \in [0, h]$. Thus, we can write the reward evaluated along the trajectory as:

$$r(x(t), u) = r(x, u) + O(t)$$

Substituting this approximation into the discounted reward integral yields:

$$\int_0^h \gamma^t r(x(t), u)\, dt = \int_0^h \gamma^t \left(r(x, u) + O(t) \right)\, dt = r(x, u) \int_0^h \gamma^t\, dt + \int_0^h \gamma^t O(t)\, dt$$

Since the discount factor $\gamma \in (0, 1)$, we have $\gamma^t \leq 1$ for all $t \geq 0$. Therefore, the error term is strictly bounded by $O(h^2)$:

[1]**Big-O**: Formally, the asymptotic notation $O(t)$ as $t \to 0$ means there exist strictly positive constants C and δ such that the norm satisfies $\|x(t) - x\| \leq C|t|$ for all $|t| < \delta$. In the context of this control system, it implies that the vector field governing the dynamics is bounded.

$$\left| \int_0^h \gamma^t O(t)\, dt \right| \leq \int_0^h 1 \cdot O(t)\, dt = O(h^2)$$

Furthermore, applying a first-order Taylor expansion to the discount factor, $\gamma^t = e^{t \ln \gamma} = 1 + O(t)$, we evaluate the main integral as:

$$\int_0^h \gamma^t\, dt = \int_0^h (1 + O(t))\, dt = h + O(h^2)$$

Combining these results, and noting that the function $r(x, u)$ is strictly bounded, we obtain the final first-order approximation:

$$\int_0^h \gamma^t r(x(t), u)\, dt = r(x, u)\left(h + O(h^2)\right) + O(h^2) = r(x, u) \cdot h + O(h^2)$$

□

Assuming the value function is continuously differentiable, $V \in C^1(\Omega)$, taking the limit of the *Dynamic Programming Principle* as $h \to 0$ reveals that V satisfies a non-linear, first-order partial differential equation known as the *Hamilton-Jacobi-Bellman (HJB)* equation.

Theorem 2.5 (*Hamilton-Jacobi-Bellman Equation*)**.** *Assume the value function V is continuously differentiable at a state $x \in \Omega$, and let $\nabla V(x)$ denote its gradient vector at x. Then, V satisfies the following non-linear partial differential equation at x:*

$$H(V, \nabla V, x) := V(x) \ln \gamma + \sup_{u \in U} \Big\{ \nabla V(x) \cdot f(x, u) + r(x, u) \Big\} = 0$$

where the supremum is taken over the instantaneous action space $U \subset \mathbb{R}^m$.

Proof. Let $u \in U$ be an arbitrary constant action applied over a small time interval $[0, h)$. By Theorem 2.3 (Dynamic Programming Principle) and Lemma 2.4, we can approximate the value function for a sufficiently small $h > 0$ as:

$$V(x) = \sup_{u \in U} \left\{ \int_0^h \gamma^t r(x(t), u)\, dt + \gamma^h V(x(h)) \right\}$$

Applying the result of Lemma 2.4 to the integral, we obtain:

$$V(x) = \sup_{u \in U} \left\{ r(x, u) \cdot h + \gamma^h V(x(h)) + o(h) \right\}$$

Next, we perform first-order Taylor expansions around $h = 0$ for the terms dependent on the system's dynamics, collecting all strictly higher-order terms into the **little-o** notation $o(h)$[2]. Since $V \in C^1(\Omega)$, the state and value function evolve as:

$$\begin{aligned} \gamma^h &= 1 + h \ln \gamma + o(h) \\ x(h) &= x + f(x, u)h + o(h) \\ V(x(h)) &= V(x) + \nabla V(x) \cdot f(x, u)h + o(h) \end{aligned}$$

Multiplying the expanded discount factor and the expanded future value function yields:

[2]Formally, the **little-o** notation $o(h)$ as $h \to 0$ represents any error function $g(h)$ that satisfies $\lim_{h \to 0} \frac{g(h)}{h} = 0$. In this derivation, it systematically absorbs all second-order and higher Taylor remainder terms, guaranteeing that they vanish completely upon dividing by h and taking the limit.

$$\begin{aligned}\gamma^h V(x(h)) &= \big(1 + h\ln\gamma + o(h)\big)\big(V(x) + \nabla V(x)\cdot f(x,u)h + o(h)\big)\\ &= V(x) + V(x)h\ln\gamma + \nabla V(x)\cdot f(x,u)h + o(h)\end{aligned}$$

Substituting this expansion back into the recursive equation gives:

$$V(x) = \sup_{u\in U}\left\{ r(x,u)h + V(x) + V(x)h\ln\gamma + \nabla V(x)\cdot f(x,u)h + o(h)\right\}$$

Because $V(x)$ and $V(x)h\ln\gamma$ depend only on the current state x and not on the chosen action u, they can be factored out of the supremum operator. Subtracting $V(x)$ from both sides of the equation, we get:

$$0 = V(x)h\ln\gamma + \sup_{u\in U}\left\{\Big(r(x,u) + \nabla V(x)\cdot f(x,u)\Big)h + o(h)\right\}$$

Finally, dividing the entire equation by h and taking the limit as $h \to 0$ eliminates the strictly higher-order terms $\lim_{h\to 0}\frac{o(h)}{h} = 0$, directly yielding the Hamilton-Jacobi-Bellman equation:

$$0 = V(x)\ln\gamma + \sup_{u\in U}\Big\{ r(x,u) + \nabla V(x)\cdot f(x,u)\Big\}$$

which is precisely $H(V, \nabla V, x) = 0$. □

The Hamilton-Jacobi-Bellman equation derived in Theorem 2.5 provides the exact continuous-time, continuous-space optimality condition. However, due to its non-linear nature, analytical solutions are generally intractable for complex dynamic systems. Consequently, we must resort to numerical approximation schemes.

To construct a computable discrete-time Markov Decision Process (MDP) that approximates the continuous control problem, we discretize the state space Ω. Let $\Sigma^\delta \subset \Omega$ be a finite grid of states (vertices) forming a simplicial complex, where $\delta > 0$ represents the maximum diameter of any simplex in the mesh.

When the system is at a grid node $\xi \in \Sigma^\delta$ and an action $u \in U$ is applied for a discrete time step $\tau = \delta$, the deterministic transition $\eta = \xi + \delta f(\xi, u)$ will generally not land on another grid node. To resolve this off-grid transition, we project the value of the target state η back onto the grid using barycentric interpolation. Specifically, if η falls within a simplex defined by vertices $\{\xi_i\}$, its value is approximated as a convex combination $\sum_i p_i W(\xi_i)$, where $p_i = p(\eta \mid \xi_i) \geq 0$ are the exact barycentric coordinates such that $\sum_i p_i = 1$ and $\sum_i p_i\xi_i = \eta$.

This spatial interpolation, combined with the temporal discretization of the Bellman Principle, yields the fundamental numerical operator of our scheme.

Definition 2.6 (Barycentric Dynamic Programming Operator)**.** Let $\Omega \subset \mathbb{R}^d$ be the continuous state space and $\Sigma^\delta \subset \overline{\Omega}$ be a finite spatial grid forming a simplicial complex, where $\delta > 0$ denotes the maximum simplex diameter. Let $W : \Sigma^\delta \to \mathbb{R}$ be a discrete value function evaluated at the grid vertices. For any grid node $\xi \in \Sigma^\delta \cap \Omega$ and any admissible action $u \in U$, let $\tau(\xi, u) > 0$ be a local time step strictly satisfying $\tau(\xi, u) = \mathcal{O}(\delta)$. The Barycentric Dynamic Programming Operator F^δ is defined as:

$$\big(F^\delta[W](\cdot)\big)(\xi) := \sup_{u\in U}\Big\{\gamma^{\tau(\xi,u)}\sum_i p(\eta \mid \xi_i)W(\xi_i) + \tau(\xi,u)\, r(\xi,u)\Big\}$$

where the discrete target state is computed via a forward Euler integration step:

$$\eta(\xi, u) = \xi + \tau(\xi, u)f(\xi, u)$$

and $p(\eta \mid \xi_i) \geq 0$ are the exact barycentric coordinates of the target state η with respect to the vertices ξ_i of its enclosing simplex. These weights uniquely satisfy the convex combination constraints:

$$\sum_i p(\eta \mid \xi_i) = 1 \quad \text{and} \quad \sum_i p(\eta \mid \xi_i)\xi_i = \eta$$

For a numerical scheme to be theoretically sound, the discrete operator must asymptotically converge to the continuous HJB equation as the mesh resolution becomes infinitely fine ($\delta \to 0$). This property is known as local consistency.

Lemma 2.7 (Contraction Mapping Property)**.** *Let $\mathcal{B}(\Sigma^\delta)$ be the Banach space of bounded discrete functions $W : \Sigma^\delta \to \mathbb{R}$ equipped with the supremum norm $\|W\|_\infty = \max_{\xi \in \Sigma^\delta} |W(\xi)|$. Assume there exists a strictly positive minimum time step $\tau_{\min} > 0$ such that $\tau(\xi, u) \geq \tau_{\min}$ for all $\xi \in \Sigma^\delta$ and $u \in U$. Then, the Barycentric Dynamic Programming Operator $F^\delta : \mathcal{B}(\Sigma^\delta) \to \mathcal{B}(\Sigma^\delta)$ is a strict λ-contraction with modulus $\lambda = \gamma^{\tau_{\min}} < 1$. That is, for any pair of functions $V, W \in \mathcal{B}(\Sigma^\delta)$:*

$$\|F^\delta[V] - F^\delta[W]\|_\infty \leq \gamma^{\tau_{\min}} \|V - W\|_\infty$$

Proof. Let V and W be two arbitrary discrete value functions in $\mathcal{B}(\Sigma^\delta)$. For any state $\xi \in \Sigma^\delta \cap \Omega$, we evaluate the absolute difference between the operator applied to V and W. Using the standard subadditivity property of the supremum, $|\sup_u f(u) - \sup_u g(u)| \leq \sup_u |f(u) - g(u)|$, we obtain: For brevity, define $\tau := \tau(\xi, u)$, $r := r(\xi, u)$ and $p_i \equiv p(\eta \mid \xi_i)$.

$$|F^\delta[V](\xi) - F^\delta[W](\xi)| = \Big|\sup_{u \in U}\Big\{\gamma^\tau \sum_i p_i V(\xi_i) + \tau r\Big\} - \sup_{u \in U}\Big\{\gamma^\tau \sum_i p_i W(\xi_i) + \tau r\Big\}\Big| \leq$$

$$\leq \sup_{u \in U}\Big|\gamma^{\tau_u} \sum_i p_i V(\xi_i) - \gamma^{\tau_u} \sum_i p_i W(\xi_i)\Big| = \sup_{u \in U}\Big\{\gamma^{\tau_u}\Big|\sum_i p_i\big(V(\xi_i) - W(\xi_i)\big)\Big|\Big\}.$$

By definition of barycentric interpolation, the weights are strictly non-negative ($p_i \geq 0$) and form a convex combination ($\sum_i p_i = 1$). Applying the triangle inequality to the finite sum yields:

$$\left|\sum_i p_i\big(V(\xi_i) - W(\xi_i)\big)\right| \leq \sum_i p_i \, |V(\xi_i) - W(\xi_i)|$$

Since the absolute difference at any localized vertex ξ_i is strictly bounded by the global supremum norm of the entire function, $|V(\xi_i) - W(\xi_i)| \leq \|V - W\|_\infty$, we can systematically bound the interpolation term:

$$\sum_i p_i \, |V(\xi_i) - W(\xi_i)| \leq \sum_i p_i \|V - W\|_\infty = \|V - W\|_\infty \underbrace{\sum_i p_i}_{=1} = \|V - W\|_\infty$$

Substituting this upper bound back into the supremum expression isolates the norm:

$$|F^\delta[V](\xi) - F^\delta[W](\xi)| \leq \sup_{u \in U}\Big\{\gamma^{\tau(\xi,u)} \|V - W\|_\infty\Big\}$$

Because the continuous discount factor $\gamma \in (0, 1)$ and the time step is strictly bounded below by $\tau_{\min} > 0$, the supremum of the exponential discount term is $\gamma^{\tau_{\min}} < 1$. Thus:

$$|F^\delta[V](\xi) - F^\delta[W](\xi)| \leq \gamma^{\tau_{\min}} \|V - W\|_\infty$$

Since this inequality holds globally for any arbitrary state $\xi \in \Sigma^\delta$, it must intrinsically hold for the maximum difference across the entire state grid. Taking the supremum over all ξ on the left side yields the fundamental contraction inequality:

$$\|F^\delta[V] - F^\delta[W]\|_\infty \leq \gamma^{\tau_{\min}} \|V - W\|_\infty$$

By the Banach Fixed-Point Theorem 4.2 (see Appendix), this strict contraction guarantees that the numerical operator F^δ admits a unique fixed point $V^\delta = F^\delta[V^\delta]$, and that the iterative sequence $V_{k+1} = F^\delta[V_k]$ converges uniformly to V^δ from any arbitrary initialization. □

Lemma 2.7 guarantees that the numerical scheme is well-posed, meaning the discrete operator F^δ inherently admits a unique fixed point V^δ. However, establishing the existence of a discrete solution is only the first step. For this numerical approximation to be theoretically sound and practically meaningful, we must ensure that it faithfully captures the original continuous control problem.

Specifically, as the discretization parameter δ approaches zero, the discrete dynamic programming operator must asymptotically recover the continuous Hamilton-Jacobi-Bellman (HJB) equation derived in Theorem 2.5. This property, known as *local consistency*, is the critical mathematical link that bridges the discrete Markov Decision Process back to the continuous physical dynamics.

Theorem 2.8 (First-Order Consistency of the Barycentric-DP Operator)**.** *Let $W \in C^2(\overline{\Omega})$ be a sufficiently smooth test function. Define the discrete operator for any state $\xi \in \Sigma^\delta \cap \Omega$ by fixing the temporal step exactly as the spatial resolution, $\tau = \delta$:*

$$\big(F^\delta[W](\cdot)\big)(\xi) := \sup_{u\in U}\Big\{\gamma^\delta \sum_i p(\eta \mid \xi_i)\, W(\xi_i) \;+\; \delta\, r(\xi,u)\Big\},$$

with the deterministic target state $\eta = \xi + \delta f(\xi,u)$, the enclosing vertices satisfying $\|\eta - \xi_i\| = O(\delta)$, and exact barycentric weights $p_i = p(\eta \mid \xi_i)$. Then, for each $\xi \in \Sigma^\delta \cap \Omega$:

$$F^\delta[W](\xi) = W(\xi) + \delta\, H\big(W, \nabla W, \xi\big) + O(\delta^2),$$

and hence, the numerical scheme strictly converges to the continuous Hamiltonian:

$$\lim_{\delta\to 0} \frac{F^\delta[W](\xi) - W(\xi)}{\delta} = H\big(W, \nabla W, \xi\big),$$

uniformly on compact subsets of Ω.

Proof. Fix an arbitrary state $\xi \in \Sigma^\delta \cap \Omega$ and action $u \in U$. Set the uniform time step $\tau = \delta$, the target state $\eta = \xi + \delta f(\xi,u)$, and let $\{\xi_i\}$ be the vertices of the enclosing simplex with respective barycentric weights p_i.

1. *Discount expansion.* Applying a second-order Taylor expansion to the exponential discount factor:

$$\gamma^\delta = e^{\delta \ln \gamma} = 1 + \delta \ln \gamma + O(\delta^2).$$

2. *Value expansion.* Since $W \in C^2(\overline{\Omega})$, expanding the test function around the origin state ξ yields:

$$W(\eta) = W\big(\xi + \delta f(\xi,u)\big) = W(\xi) + \delta \nabla W(\xi)\cdot f(\xi,u) + O(\delta^2).$$

3. *Interpolation error and linear cancellation.* To evaluate the interpolated value, we perform a second-order Taylor expansion of $W(\xi_i)$ around the exact target state η. Since $\|\xi_i - \eta\| \le \delta$, we have:

$$W(\xi_i) = W(\eta) + \nabla W(\eta)\cdot(\xi_i - \eta) + O(\delta^2).$$

Multiplying this expansion by the exact barycentric coordinates p_i and summing over all vertices of the simplex yields:

$$\sum_i p_i W(\xi_i) = \sum_i p_i \big[W(\eta) + \nabla W(\eta)\cdot(\xi_i - \eta) + O(\delta^2)\big].$$

By the fundamental geometric constraints of barycentric coordinates, we know that $\sum_i p_i = 1$ and $\sum_i p_i \xi_i = \eta$. Distributing the summation isolates the linear gradient term:

$$\sum_i p_i W(\xi_i) = W(\eta) \underbrace{\sum_i p_i}_{=1} + \nabla W(\eta) \cdot \underbrace{\Big(\sum_i p_i \xi_i - \eta \sum_i p_i\Big)}_{=\eta-\eta=0} + O(\delta^2) \underbrace{\sum_i p_i}_{=1}.$$

Because the linear term cancels out entirely, the interpolation strictly preserves second-order accuracy:

$$\sum_i p_i W(\xi_i) = W(\eta) + O(\delta^2).$$

Substituting the analytical expansion of $W(\eta)$ derived in Step 2 into this result gives:

$$\sum_i p_i W(\xi_i) = W(\xi) + \delta \nabla W(\xi) \cdot f(\xi, u) + O(\delta^2).$$

4. *Combine.* Multiplying the expanded discount factor and the exact interpolated value:

$$\begin{aligned}\gamma^\delta \sum_i p_i W(\xi_i) &= \big(1 + \delta \ln \gamma + O(\delta^2)\big)\big[W(\xi) + \delta \nabla W(\xi) \cdot f(\xi, u) + O(\delta^2)\big] \\ &= W(\xi) + \delta\big[W(\xi) \ln \gamma + \nabla W(\xi) \cdot f(\xi, u)\big] + O(\delta^2).\end{aligned}$$

Adding the scaled immediate reward $\delta r(\xi, u)$ to form the complete argument of the supremum:

$$\gamma^\delta \sum_i p_i W(\xi_i) + \delta r(\xi, u) = W(\xi) + \delta\big[W(\xi) \ln \gamma + \nabla W(\xi) \cdot f(\xi, u) + r(\xi, u)\big] + O(\delta^2).$$

Taking the supremum over $u \in U$ reconstructs the continuous Hamiltonian operator:

$$F^\delta[W](\xi) = W(\xi) + \delta\, H\big(W, \nabla W, \xi\big) + O(\delta^2).$$

5. *Conclusion.* Subtracting $W(\xi)$ from both sides and dividing by the resolution parameter δ:

$$\frac{F^\delta[W](\xi) - W(\xi)}{\delta} = H\big(W, \nabla W, \xi\big) + \frac{O(\delta^2)}{\delta}.$$

Taking the limit as $\delta \to 0$, the $O(\delta)$ remainder strictly vanishes:

$$\lim_{\delta \to 0} \frac{F^\delta[W](\xi) - W(\xi)}{\delta} = H\big(W, \nabla W, \xi\big).$$

This convergence holds uniformly on compact sets due to the uniform continuity of the second derivatives of $W \in C^2(\overline{\Omega})$. □

Summary: Bridging the Continuous and Discrete Domains. In this section, we established the fundamental theoretical framework necessary to approximate a continuous reinforcement learning problem using discrete computational methods. We began by formulating the optimal control problem within a continuous state and time domain, governed by deterministic dynamics and discounted future rewards. This continuous formulation naturally led to the Hamilton-Jacobi-Bellman (HJB) equation, which provides the exact analytical condition for optimality.

However, the inherent nonlinearity of the HJB equation renders analytical solutions intractable for most complex dynamic systems, necessitating a robust numerical approximation scheme. By discretizing the state space into a simplicial mesh and employing barycentric interpolation to project off-grid temporal transitions back onto the known grid vertices, we constructed the discrete Barycentric Dynamic Programming Operator.

The mathematical validity of this discretization rests upon two theoretical pillars. First, Lemma 2.7 proved that our numerical operator is a strict contraction mapping, thereby guaranteeing the existence and uniqueness of a discrete value function via the Banach Fixed-Point Theorem. Second, and most critically, Theorem 2.8 provided the definitive mathematical bridge connecting the discrete

and continuous regimes. By proving the first-order local consistency of the operator, we mathematically guaranteed that as the spatial and temporal resolution becomes infinitely fine ($\delta \to 0$), the discrete numerical approximation strictly and uniformly recovers the continuous HJB dynamics.

Having successfully anchored the discrete Markov Decision Process to the true physical continuous system, we can now confidently transition to the algorithmic mechanisms required to computationally solve this discretized control problem.

3. Discrete approach of the problem

Policy Iteration is a classic algorithm in Reinforcement Learning (RL) used to find the optimal policy for a Markov Decision Process (MDP). It alternates between two main steps:

(1) Policy Evaluation: Compute the value function for the current policy.
(2) Policy Improvement: Update the policy based on the computed value function.

This process continues iteratively until the policy becomes stable (i.e., no longer changes between iterations), which indicates that the optimal policy has been found.

Algorithm 1 Policy Iteration for solving MDPs

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Initialize: Arbitrarily initialize V(s) ∈ ℝ and π(s) ∈ 𝒜(s) for all s ∈ 𝒮
repeat
    Policy Evaluation:
    repeat
        Δ ← 0
        for each s ∈ 𝒮 do
            v ← V(s)
            V(s) ← Σ_{s',r} p(s', r | s, π(s)) [r + γV(s')]
            Δ ← max(Δ, |v − V(s)|)
        end for
    until Δ < θ
    Policy Improvement:
    policy-stable ← true
    for each s ∈ 𝒮 do
        a ← π(s)
        π(s) ← arg max_a Σ_{s',r} p(s', r | s, a) [r + γV(s')]
        if a ≠ π(s) then
            policy-stable ← false
        end if
    end for
until policy-stable
Return: V and π
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Theorem 3.1 (Convergence of Policy Iteration)**.** *Let $\mathcal{S}$ and $\mathcal{A}$ be finite state and action spaces. The Policy Iteration algorithm generates a sequence of policies $\{\pi_k\}$ and corresponding value functions $\{V^{\pi_k}\}$ such that $V^{\pi_{k+1}}(s) \geq V^{\pi_k}(s)$ for all $s \in \mathcal{S}$. The algorithm converges in a finite number of iterations to the optimal policy π^*, and its value function V^* is the unique fixed point of the Bellman Optimality Operator $\mathcal{T}^*$.*

Proof. Consider two consecutive policies generated by the algorithm, π_k and π_{k+1}. During the Policy Improvement step, π_{k+1} is chosen to be greedy with respect to the current value function V^{π_k}. Therefore, for any state $s \in \mathcal{S}$:

$$\sum_{s',r} p(s', r \mid s, \pi_{k+1}(s)) \left[r + \gamma V^{\pi_k}(s')\right] = \max_{a \in \mathcal{A}} \sum_{s',r} p(s', r \mid s, a) \left[r + \gamma V^{\pi_k}(s')\right] \geq V^{\pi_k}(s)$$

Let $\mathcal{T}^{\pi_{k+1}}$ denote the Bellman Expectation Operator for the new policy π_{k+1}. The component-wise inequality above can be written compactly in vector notation as:

$$\mathcal{T}^{\pi_{k+1}} V^{\pi_k} \geq V^{\pi_k}$$

To formalize the convergence, we define an iterative sequence of value functions starting from our current policy's value: let $V_0 = V^{\pi_k}$, and apply the operator recursively such that $V_{n+1} = \mathcal{T}^{\pi_{k+1}} V_n$. Since the Bellman operator is monotonic, applying it repeatedly preserves the inequality, yielding a monotonically non-decreasing sequence:

$$V_0 \leq V_1 \leq V_2 \leq \cdots \leq V_n$$

To ensure this sequence does not diverge as $n \to \infty$, we must prove it is strictly bounded above. Let $R_{\max} = \max_{s,a,s'} r$ be the maximum possible expected immediate reward. For any policy and state, the expected discounted return is strictly bounded by the infinite geometric series:

$$V_n(s) \leq \sum_{t=0}^{\infty} \gamma^t R_{\max} = \frac{R_{\max}}{1-\gamma} < \infty$$

By the Monotone Convergence Theorem, since the sequence $\{V_n\}$ is monotonically non-decreasing and strictly bounded above, its limit exists. Let us denote this limit as $V^\infty = \lim_{n\to\infty} V_n$.

Because the Bellman Expectation Operator $\mathcal{T}^{\pi_{k+1}}$ is a strict γ-contraction, it is uniformly continuous. This continuity allows us to pass the limit inside the operator's argument. Evaluating the operator at the limit yields:

$$\mathcal{T}^{\pi_{k+1}}(V^\infty) = \mathcal{T}^{\pi_{k+1}}\left(\lim_{n\to\infty} V_n\right) = \lim_{n\to\infty} \mathcal{T}^{\pi_{k+1}} V_n = \lim_{n\to\infty} V_{n+1} = V^\infty$$

This algebraic chain demonstrates that V^∞ is a fixed point of $\mathcal{T}^{\pi_{k+1}}$. By the Banach Fixed Point Theorem, this fixed point is unique and corresponds to the true value function of the new policy, meaning $V^\infty = V^{\pi_{k+1}}$. Consequently, evaluating the extremes of our initial sequence, we establish the strict improvement:

$$V^{\pi_k} = V_0 \leq \lim_{n\to\infty} V_n = V^{\pi_{k+1}}$$

Thus, the sequence of value functions $\{V^{\pi_k}\}$ generated at each algorithmic iteration is monotonically non-decreasing. Since the state space $\mathcal{S}$ and action space $\mathcal{A}$ are finite, the total number of distinct deterministic policies is exactly $|\mathcal{A}|^{|\mathcal{S}|}$. Because each policy improvement step yields a strictly better policy (until optimality is reached), the algorithm evaluates a strictly novel policy at each step. By the Pigeonhole Principle, the algorithm must terminate in at most $|\mathcal{A}|^{|\mathcal{S}|}$ iterations.

At this point of finite convergence (let us denote it as iteration N), the policy no longer changes ($\pi_{N+1} = \pi_N$), and the policy improvement equation becomes:

$$V^{\pi_N}(s) = \max_{a \in \mathcal{A}} \sum_{s',r} p(s', r \mid s, a) \left[r + \gamma V^{\pi_N}(s')\right]$$

This is precisely the Bellman Optimality Equation, meaning $V^{\pi_N} = \mathcal{T}^* V^{\pi_N}$. The resulting value function is a fixed point of the Bellman Optimality Operator $\mathcal{T}^*$. By the Banach Fixed Point Theorem, this fixed point is unique, guaranteeing that $V^{\pi_N} = V^*$ and the policy is globally optimal. □

3.1. **Hessian-based Adaptive Simplicial Mesh.**

Definition 3.2 (Hessian-Based Adaptive Simplicial Mesh)**.** Let $\Omega \subset \mathbb{R}^d$ be the continuous state space and $W \in C^2(\overline{\Omega})$ be a twice-differentiable value function. Given a uniform interpolation error tolerance $\epsilon > 0$, we define a Hessian-Based Adaptive Simplicial Mesh $\Sigma^\epsilon \subset \overline{\Omega}$ as a valid simplicial complex equipped with a state-dependent local resolution function $\delta : \Omega \to \mathbb{R}^+$.

For any state $\xi \in \Omega$, let $\delta(\xi)$ denote the maximum diameter of any simplex containing ξ. The mesh Σ^ϵ is strictly ϵ-adaptive with respect to W if its local diameter rigorously satisfies the following curvature-bounded condition:

$$\delta(\xi) \leq \sqrt{\frac{\epsilon}{\|\nabla^2 W(\xi)\| + \nu}} \quad \forall \xi \in \Omega,$$

where $\|\nabla^2 W(\xi)\|$ denotes the induced spectral norm of the Hessian matrix evaluated at the state ξ, and $\nu > 0$ is a strictly positive, user-defined regularization constant.

Definition 3.3 (Adaptive Barycentric Dynamic Programming Operator)**.** Let $\Sigma^\epsilon \subset \overline{\Omega}$ be a Hessian-based adaptive simplicial mesh with a local resolution function $\delta(\xi)$ bounded by the error tolerance $\epsilon > 0$. Let $W : \Sigma^\epsilon \to \mathbb{R}$ be a discrete value function evaluated at the adaptive grid vertices.

For any grid node $\xi \in \Sigma^\epsilon \cap \Omega$ and any admissible action $u \in U$, we define a state-dependent local time step $\tau(\xi, u) > 0$ strictly coupled to the local mesh geometry such that $\tau(\xi, u) = \mathcal{O}(\delta(\xi))$.

The Adaptive Barycentric Dynamic Programming Operator F^ϵ is defined as:

$$\big(F^\epsilon[W](\cdot)\big)(\xi) := \sup_{u \in U} \Big\{ \gamma^{\tau(\xi,u)} \sum_i p(\eta \mid \xi_i) W(\xi_i) + \tau(\xi, u)\, r(\xi, u) \Big\}$$

where the discrete target state is computed via the localized forward Euler step:

$$\eta(\xi, u) = \xi + \tau(\xi, u) f(\xi, u)$$

and $p(\eta \mid \xi_i) \geq 0$ are the exact barycentric coordinates of η with respect to the vertices ξ_i of its enclosing simplex in Σ^ϵ, uniquely satisfying the geometric constraints $\sum_i p(\eta \mid \xi_i) = 1$ and $\sum_i p(\eta \mid \xi_i)\xi_i = \eta$.

Lemma 3.4 (Adaptive Contraction Mapping Property)**.** *Let $\mathcal{B}(\Sigma^\epsilon)$ be the Banach space of bounded discrete functions $W : \Sigma^\epsilon \to \mathbb{R}$ equipped with the supremum norm $\|W\|_\infty = \max_{\xi \in \Sigma^\epsilon} |W(\xi)|$.*

Assume the continuous value function $W \in C^2(\overline{\Omega})$ has a strictly bounded Hessian such that $\sup_{\xi \in \Omega} \|\nabla^2 W(\xi)\| \leq M < \infty$. Consequently, the adaptive mesh resolution admits a strict lower bound $\delta_{\min} = \sqrt{\epsilon/(M + \nu)} > 0$, ensuring the local time step satisfies $\tau(\xi, u) \geq \tau_{\min} > 0$ globally.

Then, the Adaptive Barycentric Dynamic Programming Operator $F^\epsilon : \mathcal{B}(\Sigma^\epsilon) \to \mathcal{B}(\Sigma^\epsilon)$ is a strict contraction with modulus $\lambda = \gamma^{\tau_{\min}} < 1$. That is, for any pair of functions $V, W \in \mathcal{B}(\Sigma^\epsilon)$:

$$\|F^\epsilon[V] - F^\epsilon[W]\|_\infty \leq \gamma^{\tau_{\min}} \|V - W\|_\infty$$

Proof. Let V and W be arbitrary functions in $\mathcal{B}(\Sigma^\epsilon)$. We evaluate the absolute difference of the adaptive operator applied to these functions at any grid node $\xi \in \Sigma^\epsilon \cap \Omega$. Utilizing the subadditivity of the supremum, $|\sup f - \sup g| \leq \sup |f - g|$, we isolate the recursive terms:

$$\begin{aligned} |F^\epsilon[V](\xi) - F^\epsilon[W](\xi)| &= \left| \sup_{u \in U} \Big\{ \gamma^{\tau(\xi,u)} \sum_i p_i V(\xi_i) + \tau(\xi, u) r \Big\} - \sup_{u \in U} \Big\{ \gamma^{\tau(\xi,u)} \sum_i p_i W(\xi_i) + \tau(\xi, u) r \Big\} \right| \\ &\leq \sup_{u \in U} \left| \gamma^{\tau(\xi,u)} \sum_i p_i \big(V(\xi_i) - W(\xi_i)\big) \right|. \end{aligned}$$

Since the barycentric weights $p_i \equiv p(\eta \mid \xi_i)$ inherently satisfy $p_i \geq 0$ and $\sum_i p_i = 1$, we apply the triangle inequality to the localized interpolation sum:

$$\left| \sum_i p_i \big(V(\xi_i) - W(\xi_i)\big) \right| \leq \sum_i p_i |V(\xi_i) - W(\xi_i)| \leq \sum_i p_i \|V - W\|_\infty = \|V - W\|_\infty.$$

Substituting this uniform bound back into the supremum operator yields a state-dependent inequality:

$$|F^\epsilon[V](\xi) - F^\epsilon[W](\xi)| \leq \sup_{u \in U} \left\{ \gamma^{\tau(\xi,u)} \right\} \|V - W\|_\infty.$$

Crucially, the global bound on the Hessian norm guarantees that the spatial mesh never collapses to zero ($\delta_{\min} > 0$), which analytically enforces the strict temporal lower bound $\tau(\xi, u) \geq \tau_{\min} > 0$. Because the continuous discount factor satisfies $\gamma \in (0, 1)$, the localized discount term is monotonically bounded away from unity:

$$\gamma^{\tau(\xi,u)} \leq \gamma^{\tau_{\min}} < 1 \quad \forall \xi \in \Sigma^\epsilon,\ \forall u \in U.$$

Applying this upper bound to the inequality gives:

$$|F^\epsilon[V](\xi) - F^\epsilon[W](\xi)| \leq \gamma^{\tau_{\min}} \|V - W\|_\infty.$$

Since this bound is independent of the evaluated state ξ, it holds universally across the entire adaptive grid. Taking the supremum over all $\xi \in \Sigma^\epsilon$ fully reconstructs the spatial norm on the left-hand side:

$$\|F^\epsilon[V] - F^\epsilon[W]\|_\infty \leq \gamma^{\tau_{\min}} \|V - W\|_\infty.$$

By the Banach Fixed-Point Theorem, this strict contraction guarantees that, for any given adaptive mesh Σ^ϵ, the numerical operator F^ϵ admits a globally unique discrete optimal value function V^ϵ. □

4. Appendix

Definition 4.1 (Big-O Notation)**.** We say that $f(u, h) = O(g(h))$ uniformly in $u \in U$ if and only if there exist constants $C > 0$ and $h_0 > 0$ such that:

$$\forall u \in U,\ \forall\, 0 < h < h_0 : \quad |f(u, h)| \leq C g(h).$$

Theorem 4.2 (Banach Fixed Point Theorem)**.** *Let (X, d) be a non-empty complete metric space, and let $T : X \to X$ be a* contraction mapping*, i.e., there exists a constant $0 \leq \gamma < 1$ such that*

$$d(Tx, Ty) \leq \gamma \cdot d(x, y) \quad \textit{for all } x, y \in X.$$

Then:

1. *There exists a* ***unique fixed point*** *$x^* \in X$ such that $T(x^*) = x^*$.*
2. *For any initial point $x_0 \in X$, the sequence defined recursively by $x_{n+1} = T(x_n)$ converges to x^*.*
3. *The convergence is geometric, satisfying the error bound:*

$$d(x_n, x^*) \leq \frac{\gamma^n}{1 - \gamma} d(x_1, x_0).$$

Proof. Let $x_0 \in X$ be an arbitrary initial point, and define the iterative sequence:

$$x_1 = T(x_0), \quad x_2 = T(x_1), \quad \ldots, \quad x_{n+1} = T(x_n).$$

Step 1: Prove that $\{x_n\}$ is a Cauchy sequence. First, note the distance between consecutive terms:

$$d(x_{n+1}, x_n) = d(Tx_n, Tx_{n-1}) \leq \gamma d(x_n, x_{n-1}).$$

By induction, we can bound this distance in terms of the initial step:

$$d(x_{n+1}, x_n) \leq \gamma^n d(x_1, x_0).$$

Now, for any $m > n$, we repeatedly apply the triangle inequality:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq \sum_{k=n}^{m-1} \gamma^k d(x_1, x_0) = d(x_1, x_0) \sum_{k=n}^{m-1} \gamma^k.$$

Since $\gamma \in (0, 1)$, we can strictly bound the finite sum by the infinite geometric series:

$$\sum_{k=n}^{\infty} \gamma^k = \frac{\gamma^n}{1-\gamma}.$$

Hence, for all $m > n$, we obtain the upper bound:

$$d(x_m, x_n) \leq \frac{\gamma^n}{1-\gamma} d(x_1, x_0).$$

As $n \to \infty$, since $\gamma < 1$, the term $\gamma^n \to 0$. This implies that $d(x_m, x_n) \to 0$, which formally proves that $\{x_n\}$ is a Cauchy sequence. Because X is a complete metric space, every Cauchy sequence converges; thus, there exists an element $x^* \in X$ such that $x_n \to x^*$.

Step 2: Prove that x^* is a fixed point. Since T is a contraction mapping, it is globally Lipschitz continuous, and therefore preserves limits. We evaluate T at the limit of the sequence:

$$T(x^*) = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} T(x_n) = \lim_{n\to\infty} x_{n+1} = x^*.$$

Thus, x^* is indeed a fixed point of